

AN INFINITE FAMILY OF POLYOMINOES WHOSE C_4 -FACE-MAGIC SPECTRUM OMITTS ITS CENTRE

ABSTRACT. For a polyomino P with $n = |V(P)|$, Chalise, Low and Eisenkolb-Vaithyanathan define the C_4 -face-magic spectrum $\text{Spec}_{C_4}(P)$ to be the set of magic constants of C_4 -face-magic labelings of P , and ask (Question 2 of their Section 5) whether $\text{Spec}_{C_4}(P)$ is always an interval of integers. It is not. For the L-tromino L_2 we determine the spectrum exactly,

$$\text{Spec}_{C_4}(L_2) = \{15, 16, 17, 19, 20, 21\},$$

so the centre $2(n+1) = 18$ is missing. The mechanism is a double count: if some multiset of cells covers $V(P)$ with multiplicity one except for a single uncovered vertex and a single doubled vertex, then $4 \mid n$, the value $2(n+1)$ is excluded, and $\text{Spec}_{C_4}(P)$ is confined to the six values $2(n+1) + j$ with $1 \leq |j| \leq 3$. An explicit L-shaped family L_m , m even, satisfies the hypothesis for every m , so the centre is omitted at every order $n = 2m + 4$ with m even. Both the exclusion and the six-value window are proved without any search.

1. INTRODUCTION

A *polyomino* P is a finite edge-connected union of unit squares in the square lattice $\mathbb{Z}^2$, identified with its associated plane graph, whose vertices are the lattice vertices of the cells and whose edges are the lattice edges [1, Section 1]. We write $\mathcal{C}(P)$ for the set of cells, $V(P)$ for the vertex set, $n = |V(P)|$ and $[n] = \{1, \dots, n\}$. Throughout, the cell $[x, y]$ is the closed unit square with lower-left lattice point (x, y) , so

$$V([x, y]) = \{(x, y), (x+1, y), (x, y+1), (x+1, y+1)\}.$$

Chalise, Low and Eisenkolb-Vaithyanathan [1] prove the Shiu–Low–Liu conjecture [3] that every polyomino is C_4 -face-magic, and in their Section 5 they introduce the *C_4 -face-magic spectrum*

$$(1) \quad \begin{aligned} \text{Spec}_{C_4}(P) = \{ & s \in \mathbb{N} : \exists \text{ a bijection } \lambda : V(P) \rightarrow [n] \\ & \text{such that } \sum_{v \in V(C)} \lambda(v) = s \text{ for every } C \in \mathcal{C}(P) \}. \end{aligned}$$

They observe that $\text{Spec}_{C_4}(P)$ is nonempty for every polyomino, that the complementary labeling $\bar{\lambda} = n+1 - \lambda$ gives the symmetry

$$(2) \quad s \in \text{Spec}_{C_4}(P) \iff 4(n+1) - s \in \text{Spec}_{C_4}(P),$$

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and that $\text{Spec}_{C_4}(P) \subseteq \{10, 11, \dots, 4n-6\}$. Because of (2) the point $2(n+1)$ is the *centre* of the spectrum. They then pose five questions, the second of which reads, verbatim,

“Is $\text{Spec}_{C_4}(P)$ always an interval of integers?”

[1, Section 5, Question 2]. We answer it negatively; a witness has three cells, and the obstruction is a counting identity that needs no computer.

Theorem 1. *Let L_2 be the L -tromino, the polyomino with cells $[0, 0]$, $[0, 1]$, $[1, 0]$. Then $n = |V(L_2)| = 8$, the centre is $2(n+1) = 18$, and*

$$\text{Spec}_{C_4}(L_2) = \{15, 16, 17, 19, 20, 21\}.$$

In particular $18 \notin \text{Spec}_{C_4}(L_2)$ while 15 and 21 do lie in it, so $\text{Spec}_{C_4}(L_2)$ is not an interval of integers and Question 2 of [1] is false.

Theorem 2. *For even $m \geq 2$ let L_m be the polyomino with cells $[0, 0]$, $[0, 1]$, $\dots$, $[0, m-1]$ together with $[1, 0]$, so that L_2 is the L -tromino and L_4 the L -pentomino. Then $n = |V(L_m)| = 2m+4$,*

$$2(n+1) = 4m+10 \notin \text{Spec}_{C_4}(L_m),$$

$$\text{Spec}_{C_4}(L_m) \subseteq \{2(n+1) + j : 1 \leq |j| \leq 3\}.$$

Since $\text{Spec}_{C_4}(L_m)$ is nonempty and symmetric about $2(n+1)$, it contains values on both sides of the missing centre, so it is not an interval. Question 2 of [1] therefore fails for infinitely many polyominoes, at infinitely many orders.

Both mechanisms used below are in print and are references of [1]: Curran [2, p. 362] confines the C_4 -face-magic value of an odd-order projective grid graph $P_{m,n}$ to $\{2mn+1, 2mn+2, 2mn+3\}$ by the same kind of double count, and the label-collision device is Lemma 2.1 of Shiu–Low–Liu [3].

2. TWO LEMMAS

Lemma 3. *Let P be a polyomino with $n = |V(P)|$, and suppose there is a finite multiset S of cells of P and two distinct vertices $p, v \in V(P)$ such that, writing $w(u)$ for the number of members of S whose vertex set contains u ,*

$$w(p) = 0, \quad w(v) = 2, \quad w(u) = 1 \text{ for all } u \in V(P) \setminus \{p, v\}.$$

Call such an S a single-defect cover of P . Then

- (i) $4 \mid n$ and $|S| = n/4$;
- (ii) *every $s \in \text{Spec}_{C_4}(P)$ satisfies $s = 2(n+1) + \frac{4\delta}{n}$ for some integer δ with $1 \leq |\delta| \leq n-1$;*
- (iii) $2(n+1) \notin \text{Spec}_{C_4}(P)$.

Proof. Each cell has exactly four vertices, so $\sum_{u \in V(P)} w(u) = 4|S|$. On the other hand the hypothesis gives $\sum_u w(u) = 0 + 2 + (n-2) = n$. Hence $4|S| = n$, which is (i).

Let λ be a C_4 -face-magic labeling of P with magic constant s , and put $N = \sum_{i=1}^n i = n(n+1)/2$. Summing the magic condition over the members of S and exchanging the order of summation,

$$|S|s = \sum_{C \in S} \sum_{u \in V(C)} \lambda(u) = \sum_{u \in V(P)} w(u) \lambda(u) = N - \lambda(p) + \lambda(v).$$

Writing $\delta = \lambda(v) - \lambda(p)$ and $|S| = n/4$ this is $(n/4)s = N + \delta$, i.e.

$$(3) \quad s = \frac{4(N + \delta)}{n} = 2(n+1) + \frac{4\delta}{n}.$$

Since λ is a bijection onto $[n]$ and $p \neq v$, we have $\lambda(p) \neq \lambda(v)$, so $\delta \neq 0$, and both labels lie in $[n]$, so $|\delta| \leq n-1$. That is (ii).

For (iii), take $s = 2(n+1)$ in (3): it forces $\delta = 0$, i.e. $\lambda(p) = \lambda(v)$ with $p \neq v$, contradicting injectivity. Equivalently and with no algebra: $(n/4) \cdot 2(n+1) = n(n+1)/2 = N$, so the identity reads $N = N - \lambda(p) + \lambda(v)$. $\square$

Lemma 4. *Under the hypothesis of Lemma 3, write $n = 4t$. Then*

$$\text{Spec}_{C_4}(P) \subseteq \{2(n+1) + j : j \in \{-3, -2, -1, 1, 2, 3\}\},$$

so $|\text{Spec}_{C_4}(P)| \leq 6$, however large P is.

Proof. By (3), $s - 2(n+1) = 4\delta/n = \delta/t$. As s is an integer, $t \mid \delta$, say $\delta = jt$ with $j \in \mathbb{Z}$. By Lemma 3(ii), $\delta \neq 0$ and $|\delta| \leq n-1 = 4t-1$, so $0 < |j| \leq 4 - 1/t < 4$ and hence $j \in \{\pm 1, \pm 2, \pm 3\}$. Finally $s = 2(n+1) + j$. $\square$

So a polyomino with a single-defect cover has a spectrum contained in the six integers within distance 3 of its centre, and missing the centre itself. Nothing but a covering count has been used; in particular no labeling has been searched for.

3. THE L-TROMINO

Proof of Theorem 1. The cells of L_2 are $[0, 0]$, $[0, 1]$, $[1, 0]$, so

$$V(L_2) = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1)\},$$

$n = 8$, $t = n/4 = 2$ and the centre is $2(n+1) = 18$.

Upper bound. Take $S = \{[0, 1], [1, 0]\}$. Its two cells have vertex sets $\{(0, 1), (1, 1), (0, 2), (1, 2)\}$ and $\{(1, 0), (2, 0), (1, 1), (2, 1)\}$, so $(1, 1)$ is covered twice, $(0, 0)$ is covered not at all, and each of the remaining five vertices exactly once. Thus S is a single-defect cover with $p = (0, 0)$, $v = (1, 1)$ and $|S| = 2 = n/4$. By Lemmas 3 and 4,

$$\text{Spec}_{C_4}(L_2) \subseteq \{15, 16, 17, 19, 20, 21\}.$$

Lower bound. Each of the six columns of Table 1 is a bijection $V(L_2) \rightarrow [8]$ under which all three cells of L_2 sum to the value heading the column. Hence each of the six values lies in $\text{Spec}_{C_4}(L_2)$, and the two bounds meet.

Since $15, 21 \in \text{Spec}_{C_4}(L_2)$ and $18 \notin \text{Spec}_{C_4}(L_2)$, the set is not an interval. $\square$

u	$\lambda_{15}(u)$	$\lambda_{16}(u)$	$\lambda_{17}(u)$	$\lambda_{19}(u)$	$\lambda_{20}(u)$	$\lambda_{21}(u)$
(0, 0)	7	5	3	1	1	1
(0, 1)	5	8	8	8	8	8
(0, 2)	3	3	2	2	3	2
(1, 0)	2	2	5	7	6	5
(1, 1)	1	1	1	3	5	7
(1, 2)	6	4	6	6	4	4
(2, 0)	4	6	4	4	2	3
(2, 1)	8	7	7	5	7	6

TABLE 1. Six C_4 -face-magic labelings of the L-tromino L_2 , one for each value of $\text{Spec}_{C_4}(L_2) = \{15, 16, 17, 19, 20, 21\}$; λ_s has magic constant s . For example λ_{17} gives cell $[0, 0]$ the sum $3 + 5 + 8 + 1 = 17$, cell $[0, 1]$ the sum $8 + 1 + 2 + 6 = 17$ and cell $[1, 0]$ the sum $5 + 4 + 1 + 7 = 17$.

4. THE INFINITE FAMILY

Proof of Theorem 2. Fix an even $m \geq 2$. The m column cells $[0, j]$, $0 \leq j \leq m-1$, together contribute the $2(m+1)$ vertices $\{0, 1\} \times \{0, \dots, m\}$, and the cell $[1, 0]$ contributes in addition the two vertices $(2, 0)$ and $(2, 1)$. Hence

$$V(L_m) = (\{0, 1\} \times \{0, \dots, m\}) \cup \{(2, 0), (2, 1)\},$$

$$n = 2m + 4.$$

As m is even, $4 \mid n$.

Take

$$S = \{[0, 1], [0, 3], \dots, [0, m-1]\} \cup \{[1, 0]\}, \quad |S| = \frac{m}{2} + 1 = \frac{n}{4}.$$

The cell $[0, 2i-1]$ has vertex set $\{0, 1\} \times \{2i-1, 2i\}$, and as i runs over $1, \dots, m/2$ these sets partition $\{0, 1\} \times \{1, \dots, m\}$; so every vertex of the column except $(0, 0)$ and $(1, 0)$ is covered exactly once. The cell $[1, 0]$ covers $(1, 0), (2, 0), (1, 1), (2, 1)$. Of these, $(2, 0)$ and $(2, 1)$ were uncovered and become covered once, $(1, 0)$ was uncovered and becomes covered once, and $(1, 1)$ was already covered once and becomes covered twice. The vertex $(0, 0)$ is covered by no member of S . Thus S is a single-defect cover with $p = (0, 0)$ and $v = (1, 1)$, and Lemmas 3 and 4 give both displayed assertions of the theorem.

That $\text{Spec}_{C_4}(L_m)$ is nonempty is the main theorem of [1], and (2) makes it symmetric about $2(n+1)$; so if s_0 is any element then so is $4(n+1) - s_0$, and the two lie strictly on opposite sides of the centre, which is absent. Hence $\min < 2(n+1) < \max$ with the centre missing, and the set is not an interval. $\square$

Parity is forced rather than chosen: $|S| = n/4$ must be an integer, and $n = 2m+4$ is divisible by 4 exactly when m is even. For odd m the argument above says nothing whatever, and we make no claim there.

Not claimed. That all six window values are actually attained is proved here only for L_2 , by the labelings of Table 1; for larger even m we assert only nonemptiness, which is [1], together with the six-value window. Nothing here shows that L_2 is a polyomino of least order with a non-interval spectrum, nor that non-interval spectra arise only from single-defect covers, nor that the gap must always be a single value at the centre. Question 3 of [1, Section 5], which asks for a characterisation of the polyominoes P with $2(n+1) \in \text{Spec}_{C_4}(P)$, remains open: Lemma 3(iii) gives a sufficient condition for failure, not a necessary one.

5. VERIFICATION

Everything above can be checked by hand: one adds four numbers per cell in Table 1 and checks the covering multiplicities of $\{[0, 1], [1, 0]\}$ and of the family cover against the vertex lists.

A program `verify.py` accompanies this paper, together with a recorded run `verify.output.txt`. It is Python 3.9+, standard library only, with no external data file: the cell lists, covering sets and labelings are literals in the source, transcribed from the paper. It re-derives $V(P)$ and n from the cell lists, verifies that each printed labeling is a bijection onto $[n]$ and that every cell sums to the declared constant, recomputes the covering multiplicities and the resulting values of δ , checks the arithmetic identity $(n/4) \cdot 2(n+1) = n(n+1)/2$ that forces the collision at the centre, confirms that the realised spectrum coincides with the six-value window of Lemma 4 and is symmetric under (2), and re-checks the covering rule of Theorem 2 for all one hundred even m with $2 \leq m \leq 200$. Its recorded run also checks labelings and covering profiles of further shapes that this note does not discuss. All arithmetic is exact integer arithmetic; there is no search for labelings anywhere, no floating point and no randomness, and the run takes well under a second. It reports 99 checks, all passing, and prints its own list of what it does not cover.

REFERENCES

- [1] P. Chalise, R. M. Low, and A. Eisenkolb-Vaithyanathan, *All polyominoes are C_4 -face-magic*, preprint, submitted 9 August 2026. arXiv:2608.08914v1. Section, definition and question numbers are quoted from v1 of that e-print, the only version available to us; Question 2 is reproduced verbatim in Section 1 above, so nothing here depends on the numbering.
- [2] M. J. Curran, *C_4 -face-magic labelings on odd order projective grid graphs*, Australas. J. Combin. **83** (2022), no. 3, 361–396.
- [3] W. C. Shiu, R. M. Low, and A. K. Liu, *Face-magic labelings of polygonal graphs*, Theory Appl. Graphs **11** (2024), no. 1, Art. 7. doi:10.20429/tag.2024.110107.